Elie-Joe Farah

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Education

BE in Computer Engineering, Dean's Honor List — Lebanese American University *2021–2025*
Master en Ingénieur Civil en Informatique — Université Libre de Bruxelles *2025–present*
Thesis: Price signals for shifting the electricity demand of electrified Belgian dwellings

- Designing a price-signal algorithm that steers heat pumps, water heaters and EV chargers in 500 simulated Belgian dwellings toward periods of high renewable output, accounting for forecast uncertainty.
- Built an experiment harness and an emissions pipeline using ENTSO-E generation-mix data and sourced emission factors to measure grid carbon intensity across the dwelling fleet.

Publications

Integrating Geographic Data into Life Cycle Assessment: A Spatial Analysis of European Wind Turbines
Dominik Huber, **Elie Joe Farah**, Maeva Lavigne Philippot, Luis Ramirez Camargo, Thierry Coosemans, Maarten Messagie
International Journal of Life Cycle Assessment (IJLCA) *2025*

Professional Experience

Student Researcher (Software Development) — *MOBI, EVERGi, VUB Brussels — Sep 2026 – present*

- Built a Python validation library and Streamlit web app that catch malformed scenario files for EVERGi's Tabletop energy simulator.
- Built a pipeline that turns energy optimisation model output into validated Tabletop scenarios.
- Deployed the tools on a VUB VM with Docker and systemd, and worked with researchers and the external vendor on data formats and bug reports.

QA Intern — *FOO* *Beirut — Jun 2025 – Jul 2025*

- Executed 2,500+ mobile/web test cases, contributing to early detection of regressions.
- Reported 25+ critical bugs via Jira, improving workflow clarity and test coverage.

Software Development and QA Intern — *ITG - ITB* *Beirut — Jun 2024 – Aug 2024*

- Built a PHP Laravel backend demo with REST-style API structuring and authentication workflows.
- Performed data-driven testing, identifying 20+ major issues and improving system stability.

Technical Projects

BatteryLCATool – Multi-Tenant LCA Data Platform | *Supabase, PostgreSQL, FastAPI, React, Python, Docker*

- Built a multi-tenant platform for battery energy systems life-cycle inventory modeling, supporting manual data entry and structured Excel import with revision control and role-based access via Supabase Auth and PostgreSQL Row-Level Security.
- Implemented an async FastAPI pipeline that parses three Excel formats, atomically persists process and exchange data, and runs a parametric validation engine with per-exchange issue tracking.
- Built a Monte Carlo simulation engine (NumPy/SciPy) sampling parameterized uncertainty distributions over energy system variables with Spearman sensitivity analysis; designed an admin flow-mapping workflow (trigram search) and Excel export pipeline with Supabase Storage delivery.
- Deployed to production on Supabase Cloud, Railway (containerized FastAPI) and Vercel with GitHub Actions CI; now migrating the full stack to a self-hosted VUB VM (Docker Compose, self-hosted Supabase, reverse proxy with TLS).

Multi-Step Forecasting of Nonlinear Dynamical Systems (NARX) | *Python, PyTorch, scikit-learn*

- Built a system identification pipeline for a two-output nonlinear (NARX) system: a wrapper-based grid search with time-series cross-validation selects the lag structure, checked against simulated systems with known dynamics.
- Implemented and compared Recursive, Direct, DirRec and DIRMO multi-step forecasting strategies, and tuned Gradient Boosting and Ridge baselines inside a scikit-learn pipeline with custom scaling and predictor steps that avoid data leakage.
- Built the final model, a PyTorch MLP ensemble that forecasts both outputs together using the recursive strategy, trained with AdamW, cosine annealing, Gaussian input-noise injection to limit error build-up, and random-search hyperparameter tuning.

AI-Aided Course Platform | *Python, SQL, LangChain, OpenAI API, React, Node.js*

- Built an AI assistant enabling semantic querying of course materials using LangChain and OpenAI APIs.
- Designed a full-stack system integrating backend APIs, SQL storage, and a React frontend, achieving 90%+ response relevance in peer testing.

PostgreSQL DNA Sequence Extension | *C, PostgreSQL, SP-GiST*

- Developed custom DNA and k-mer data types with C-based functions for sequence processing.
- Designed a trie-structured SP-GiST index enabling scalable k-mer queries.

yaLcc Compiler and LL(1) Parser | *Java, JFlex, Compiler Design*

- Implemented an LL(1) parser with grammar normalization and parsing-table generation.
- Integrated a JFlex scanner and produced LaTeX parse-tree visualizations for debugging.

Skills

Programming Languages: Python, Java, C/C++, JavaScript

Databases: PostgreSQL, SQL

Frameworks & Tools: FastAPI, React, Supabase, Docker, NumPy/SciPy

Languages: English, Arabic, French

Volunteering

- **My School Pulse**, 2018–2019 — Supported educational activities for hospitalized children.
- **Heartbeat – La Chaine de l'Espoir**, 2019–2020 — Assisted efforts to expand access to healthcare.